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Unified Framework for Effective Firing of New Generation ATGMs from Smart ICV/ Tank with NavIC and Satellite / Drone Image support

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ABSTRACT

The recent geo-political developments taking place in the field of nations' security is demanding technologically advanced ammunition. Smart new-generation anti-tank guided missiles (NG-ATGMs) with high-precision, optimal propellant with maximum range coverage, high-impact-kill for nations' security is becoming need of the hour. Quality of such smart ATGMs is also becoming a challenging task, needing careful planning of test coverages from functional, operational, electrical and chemical properties. To ensure the qualitative and quantitative test parameters clinging near to their mean values, needs quality in design, development, prototyping and its production phases. Success-rate of the firing of NG-ATGMs also depends on firing platform such as infantry combat vehicle (ICV) or a tank. This is necessitating that an ICV / tank also need to be smart enough with advanced VLSI and FPGA / System-on-Chip (SOC) based systems to give the advantage for successful and reliable firing-rate of NG-ATGM. Additionally, the possibility to utilize AI- ML techniques will further help to improve the accuracy and reliability of firing and enhance the ease of operations for the user / operator. AI-ML enabled smart ICVs will further increase the reliability of the ATGM firing. This paper brings out a potential approach to NG-ATGM that are fired from a smart ICV enabled with satellite / drone support and AI-ML for increased accuracy and reliability. The challenges & opportunities of smart NG-ATGMs firing from futuristic smart ICV with real-time-classified images and navigational support by satellites have been described. The same can be extended to tanks also. The benefits of the study are encouraging.

Keywords: New generation ATGMs; Smart ICV; Effective-firing; Machine-to-machine technology; NavIC; Smart small satellite; Field programmable gate arrays

NOMENCLATURE

NG-ATGM	: New Generation Anti-Tank Guided Missile
TBA	: Tactical Battle Area
TCP-IP	: Transport Control Protocol-Internet Protocol
AI-ML	: Artificial Intelligence and Machine Learning
FPGA	: Field Programmable Gae Array
ICV	: Infantry Combat Vehicle
IRNSS	: Indian Regional Navigational Satellite System
PL	: Programmable Logic (section of FPGA)
M2M	: Machine to Machine technology
SoC-IC	: System On Chip- Integrated Chip
QA	: Quality Assurance
CAN	: Controlled Area Net-working
SSD	: Solid State Disk
MTBF	: Mean Time Between Failures
MTTR	: Mean Time To Repair
APLAC / ILAC	: Asia Pacific Lab Acquisition Council / International Lab Acquisition Council
CNN/ANN	: Convolutional Neural Network/ Artificial Neural Network
IMEI No.	: International Mobile Equipment Identity number
BITE	: Built-In Test Equipment (Test facility)
NavIC	: Navigational Indian Constellation
CDMA	: Code Division Multiple Access
FSM	: Finite State Machine
CAN	: Controlled Area Network
SAR	: Synthetic Aperture Radar
ASIC	: Application Specific Integrated Circuit
VOIP	: Voice Over Internet Protocol
IDE	: Integrated Development Environment
NC / NO	: Normally Closed / Normally Open

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1. INTRODUCTION

Anti-Tank Guided Missiles (ATGM) are the low-cost high-precision missiles spanning over a maximum range varying from 4000-8000 mtr range either wire-guided / seeker-guided. ATGMs are fired from various platforms like Infantry Combat Vehicles (ICV), trucks, container-vehicles, or from ground-launcher platforms. DRDO has given major thrust towards ATGM development[1-2]. Evolution of various generations of ATGMs and their internal sub systems, variants and the possible exploitations are described in the research works of[3-5]. First generation ATGMs were of manually operated & guided. In 2nd generation they were upgraded to semi automatic command. The 3rd generation ATGM were enhanced to fire and forget nature. The 4th generation ATGMs were enhanced fire and forget with long range (lock-on after launch). In the 5th generation ATGM, range has been extended with computer vision models upto 50 km, as described in the works of[6]. In the recent research developments taking place in India and globally, ATGMs can also be fired from tank platform. Our research considers tank also as one of the potential platforms for ATGM firings, as the technology matures in near future.

The internals of design and development of propulsion sub system i.e., booster, sustainer, nozzle, lining and grain for ATGMs discussed in the research works[7] gave scope for adaptable systems. Active research carried out for target acquisition and guiding the ATGM discussed in the works of[8-10]. Authors in their research describe enhancement in quality and performance of ATGM[11]. Targets which need to be intercepted by ATGM have been modelled in the works of[12-13]. Efforts have been made by various researchers for tracking, using semi-automatic command line-of-sight based on digital-image-processing[14], and reduced human effort in optical tracking of targets[15] and also for modelling the optimal trajectory[16].

Research progressing contemporarily is driving the user or operator's requirements towards futuristic NG-ATGM for improved success-rate of firing from a smart ICV enabled with target detection & identification, tracking of the moving target and guidance of the ATGM with real-time satellite images and navigational guidance with NavIC. Smart ICV enabled with state-of-the-art electromechanical systems makes efficient time responsiveness in its internal operations makes ease in pre-conditioning for ATGM firing. Adding AI-ML with hybrid control system increases the accuracy / precision of internal pre-conditioning and also helps in aligning the azimuth and elevation system of firing mechanism / system. AI-ML also enables the ICV to move in a precise manner to a particular location within minimum timelines with the co-ordinates support given by NavIC. Direct delivery of object's classified images from satellite / drone enhances safety to user / operator and helps for his first firing opportunity. The ICV enabled with various pressure / wind velocity sensors locally estimates the pressure, wind speed & direction etc. Details of the TBA forms as input to the AI-ML working in the FPGA / computer sitting in the ICV. This enhances the operator with precise accurate final operation of firing at his ease. Any improvement in design, production, QA and maintenance activities enhances user's / operator's ease, safety and timely readiness for taking up the firing action.

The requirements of the tactical battle area (TBA) are demanding[17] the following parameters,

- High-impact-kill power,
- Optimum use of propellant,
- Maximum-range cover,
- High-altitude applications, high-velocity (high-Mach)/ shorter flight time,
- High-precision,
- High-accuracy,
- Mission-time reliability,
- Fire and forget,
- Local / global navigation guidance,
- Cheaper costs,
- Connectivity with M2M smart systems.

These requirements also emphasise that consistent quality is essential from the initial stages of ATGM design, development stage followed by prototyping, evaluation and its bulk production phases. These phases are succeeded by quality assurance (QA) checks further extended to the phase of delivery to end-user by safe transportation means. After receiving the delivery of the ATGMs at user location, their storage under controlled environment also contributes towards quality of the ATGM. QA aspects are also applicable to the smart ICV supported by NavIC and satellite / drone images.

In this paper a new top-down approach of design is presented.

Level-1: NG-ATGM, with built-in electronics with possible least RCS, with optimal propulsion material / grade, and high-impact-kill explosive material grade use for its manufacture by production agency.

Level-2: The traditional ICV design improvement with smart solid-state electronic sub systems and in-built high-performance Hard / soft processor that enables to compute firing angle to generate high-impact, taking into consideration of the environmental conditions of TBA has been proposed.

Level-3: Providing Satellite's onboard processed real-time direct-classified images of the works[18-20] of TBA, along-with NavIC system-based tracking & navigational guidance to smart ICV.

Level-4: The user / operator's console panel enabled with the parameters of the ICV subsystems and quick decision management feasibility for enabling to operate the firing systems.

The re-design / improvement methodology for a conventional ICV to a smart and intelligent ICV with the help of state-of-the-art solid-state electronics has been presented. A comparative study based on the qualitative parameters / attributes has been made, and the results are discussed.

Designing the smart electronic systems of the ICV / variants for user friendliness result in automation, accuracy, reliability, flexibility, upgradability, testability, productivity and quality.

In the advent of M2M (Machine to Machine) technology raising across the globe in next 5 to 10 years, the nation's security systems also need to be migrated to M2M. All ICVs' brought under M2M approach, will enable safety against self-destructions, by having secure handshaking of intercommunication. In this process, unique hard IMEI number for each sub-assembly / sub-system / ICV can be allocated. Additionally, a soft coded IMEI number with TCP-IP model of interconnection / communication helps ICV and their population can communicate or ping among each other with its peers in vicinity. This mechanism can also help to monitor 4 tier levels (Corps / Division / Brigade / Regiment), and to ping across various other Corps as is needed in the TBA.

In this paper Section 2 brings out challenges of NG-ATGM. Section 3 describes the proposed innovation in sub systems for smart ICV with NG-ATGMS. Section 4 describes innovative design of ICV management and new approach of interconnecting smart sub systems with CAN controller based micro controllers. This enables a network similar to the wireless sensor network described in the works of[21-22]. Section 5 describes NavIC and smart small satellite support for direct delivery of images for near real time autonomous[23-25] firing of NG-ATGM. Section 6 describes importance of AI-ML approach for enhanced automation to improve firing-rate of ATGMs. Section 7 gives the conclusion.

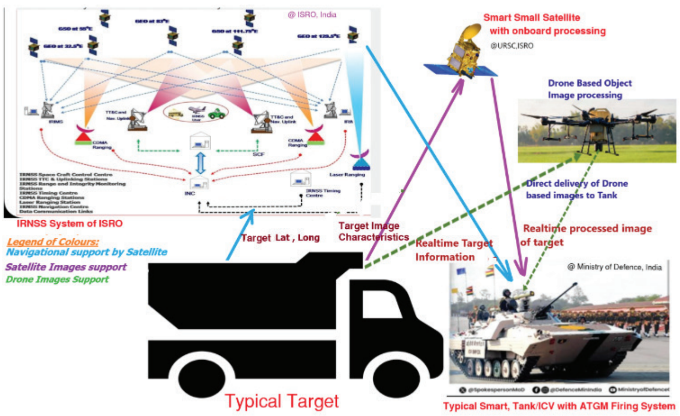

Figure 1. Smart ICV with real-time target tracking, and target image characteristics using satellite and drone support for effective smart firing of ATGM (small satellite and NavIC satellite image courtesy @ISRO, and ICV image courtesy @ Ministry of Defence official website).

2. NEW GENERATION ATGMS AND CHALLENGES

User requires optimal firing of ATGMs with ease from the platform of his choice. Operator / user may also like to perform firing of ATGMs at TBAs of various altitudes with optimal impact & fire power in all terrain purposes. Towards this, increasing potential requirements of the end-user viz.,

- Extended range of operation,
- Firing at higher altitudes,
- Firing from different platforms,
- Operator's ease of firing,
- Reliability of successful firing,
- Precision & accuracy in all weather conditions,
- Time efficient,
- Remote firing options.

All the stakeholders (i.e., designer, production agency, quality assurance agency, maintenance agency and user) need to be in synchronism for sustained-reliability for NG-ATGM firing. This approach needs to be adhered in all of the activities connected with the aspects of design, development, production and QA of NG-ATGMs.

Figure 2. Expected features of NG-ATGM fired from smart ICV's with satellite support.

Designers primarily need to ensure / facilitate recording facility for critical & important parameters of the NG-ATGMs in electronic memory system for quick retrieval. This helps for ensuring better traceability. This feature needs to be applied through-out the life cycle activities. Any updation need to be done with the involvement of all stake holders and agencies.

Minimal modifications are needed to be undertaken under careful review from technical and administrative perspectives by all the stakeholders. All the stakeholders should evaluate, agree-to / ratify at a common platform / forum for every instance of the mile-stone in ATGMs related technical activity / improvement.

Designer's established system needs to be meticulously followed / adhered by the production agency for all the NG-ATGMs under production. Each ATGM is allocated with hard coded serial number with batch and manufacturing year vis-à-vis its parameters during various stages of production / final assembly.

The concerned test-jigs and automated-test-equipment (ATE), including Go & No-Go gauges-based test results need to be recorded in a safe repository enabled with block-chain technology-based instantiation system. This ensures transparency of the test results. This helps for analysis of the trend of the ATGM firings and during the defect investigation. For ensuring accuracy, ATE / computerized measurement / recording system need to be calibrated at regular intervals. Production agency needs to maintain their test equipment as per APLAC / ILAC lab standards.

The QA agency to have a set of designers evaluated standardized ATE / computerized testing system / Go, No-Go testing gauges. This testing equipment is regularly calibrated to ensure that traceability of the measured results, is within tolerable limits as the ATGMs initially designed against specified parameters.

The maintenance agency also needs to utilize the evaluated and standardized testing equipment for assessing the

MTBF calculations and reliability prediction using the ANN based analysis between various parameters of the sub systems of the NG-ATGMs.

All the stakeholders need to have a common platform to bring out various input parameters and expected output parameters so that the observed parameters can be listed out relevantly. Additionally, the prior data from user's experiences should be collected / collated scientifically analysed using statistical tools & techniques and AI-ML techniques[25-26] for drawing inferences and suggesting prediction models for ensuring "Zero-Defect" for end user. This approach can help M2M technology-enabled cyber physical systems supported with AI-ML with established / evaluated calibrated reliability for ease of precision-firing by the nearest located ICV at the forefront of the TBA.

User or operator's tacit experiences need to be converted into check lists. These check lists once converted as inputs for CNN or ANN will serve as a "Poka-Yoke" built into the firing system for different platforms. This mechanism once evaluated by stakeholders will form as a standard system for improved effective firing. This is possible in electronics hardware by the way of employing micro switches, with their NC / NO contacts coupled to the micro controller that measures health state of ICV sub systems. This system can be more effectively evolved by applying the techniques of state machine approach for both electronic hardware and software used in the ICV.

For ensuring maximum firing range and maximum altitudes of operation designer to take into consideration of parameters viz., optimal propellant material, the L/D of the ATGM, shape of the ATGM and its nose length, material and mass of the ATGM, shape of wings, and fins complying to aerodynamics, the altitude of operation and the oxygen availability. Additionally, nozzle diameter adjustment and adaptable nozzle diameter technology helps the user with maximum range cover and effective & impactful firing of NG-ATGM. Designer also needs to consider the environmental conditions prevailing at such altitudes, viz. the velocity of winds, ambient temperature, amount of oxygen present at such altitudes, rain-fall / snow-fall. These need to be modelled as they will affect / contribute for the Mach of the NG-ATGM.

For achieving the optimal kill in different TBAs as well as dynamically varying day-to-day environmental conditions the designer need to consider the prevailing environmental parameters viz. oxygen, temperature and altitude conditions, rainfall / snowfall etc. Hence, choosing respective grade and combination of the propellant and explosive / warhead material grains enable achieving its intended purpose.

3. PROPOSED INNOVATIONS IN SUB SYSTEMS FOR SMART ICV FOR NG-ATGMS

Traditional ICV / consists of various electronic sub systems, viz., power-supply, stabilizer, communication, firing system (missile launcher, small gun etc.,), navigation, turret and hull electrical system as shown in Fig 3. Traditional analog-sub-system-based design requires considerable stabilization time, once after they are powered on. This restricts the main system's overall response time, before exploiting its operational use at TBA.

The recent innovations in semiconductor technology, and VLSI technology in the field of electronics are giving feasibility for traditional analogue based sub systems to be evolved as digital sub-systems. Few systems, tend to continue as analogue, for various legacy advantages. However, measurement of various parameters of these analogue sub systems, always remains as a primary concern / objective from end user point of view.

This necessitates, ease and accurate measurement of the health of the sub systems including the enhanced firing capability of the ICV using the ATGMs. Innovative approaches to traditional ICVs' are proposed for automated measurement and display of the parameters on the dash board of the operator, to enable user to focus for ease of exploiting & performing successful operations. Implementation of each of the electronic sub systems with FPGA / micro controller or multiprocessor-based systems will yield re-configurability or retro-modification.

The overall health of the electronic system is retrieved from all individual smart electronic sub systems and is monitored on the state-of-the-art processor with ruggedized touch-screen display for operational ease of the user / gunner and commander and troopers. This further facilitates integrated operating & controlled environment, while the ICV is under move or under stand-still condition. Redundancy built into these electronics subsystems designed with FPGA / micro controller ensures as fault-tolerant & reliable, thus largest meant time between failures (MTBF), with least MTTR.

This novel approach to re-orientation of design of electronic sub systems contributes for automation of ICV. This facilitates agile design, improved production, assembling, testing and operational aspects. The micro controller-based design is described, and the connected sub systems' reorientation into smart sub systems are described.

Re-orientation of the conventional sub systems to the feasible extent, will yield transparency of the Health of the ICV and the ATGM firing system and all the other systems test parameters with traceability. This ensures user's confidence in

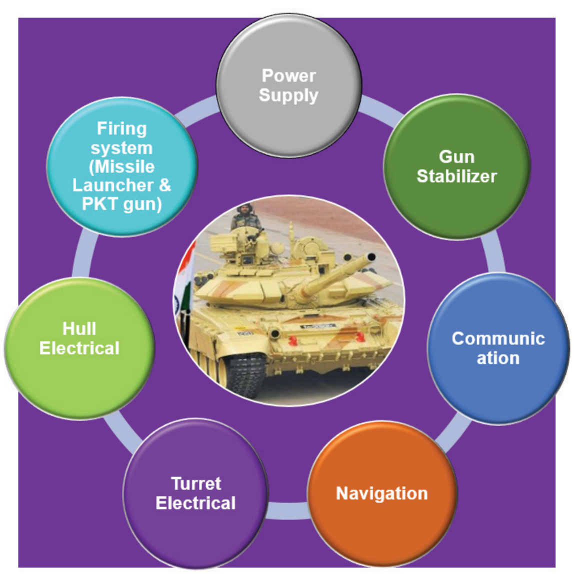

Figure 3. ICV sub-systems (image courtesy from Ministry of Defence website and Indian-express paper).

exploiting the ICV as well as implicit compliant to Industry 4.0 / 5.0 manner of recording the vital parameters of all the electronic sub systems, in the SSD (Solid-State Recorder Memory) recorder.

3.1 Power Supply System

Power measurement sensors coupled with micro controller or FPGA in the Power sub system of ICV helps to capture power consumption taking place at regular intervals. Power supply system designed with solid-state power measurement sensors, digitally controllable signal & power supply bus, with feedback system controls for optimal usage of power stored in smart batteries and will minimise weight & volume. This increases overall efficiency of the firing platform. The smart / intelligent rectifiers coupled to AC power derived from power-take-off from ICV or from inverters charge the intelligent / smart battery mechanism. In the back-end, processing of power-budget and its allocation to sub systems with a rational-algorithm implemented in the hard-processor or a soft processor is possible for implementation using FPGA / micro controller / system-on-chip hardware for dynamic power budget management purposes.

Smart-grid model (scaling down for its use in ICV) of power management will be suitably adapted for dynamic allocation of the power with power management being done with micro-controller / FPGAs based hardware logic. This approach in the ICV/tank helps automated dynamic power allocation / management system21. Considering the power generation through alternator helps optimal power allocation to all sub systems of ICV. Power supplied by smart batteries which are getting charged while the ICV is under mobility assisted by micro-controller or FPGA. Self-calibration or BITE facility by connecting outputs of all the associated sensors, will enable user to diagnose the inherent faults if any before proceeding for carrying out operations. Figure 4 shows the Smart Power supply system.

3.2 Stabiliser

Stabilizer is a critical system that firmly aligns the gun or missile launcher pointing towards target and enables the operator to fire the ammunition. Positional inputs (azimuth and elevation data) from electronic encoder / gyros of the ICV are fed to micro controller system for intercomparison with the aiming gun and target's reference co-ordinates. Corrections that are necessary, are generated by closed loop feedback system and supplied to controlling devices i.e., azimuth & elevation (X and Y) drives, so that precise gun alignment is achieved and stabilized firmly aiming to the target, for conduct of ATGM firing.

The proposed stabilizer system implementation with an FPGA or micro controller having computational resources (power), memory size, and effective computing algorithms implemented in the PL section results a high-performance hard processor with low latency. The computing hard processor of PL section obtains the information from the target tracking sensors locally, and compares it with database or by rule-governed pattern classifiers. Speed radar sensor helps in sensing the target velocity and enables the back-end processor for accurate estimates. Weather sensor / wind profiler data, altitude sensor data, and oxygen content in the TBA are collected, for computing the projectile at which ATGM needs to be fired. This helps to align the gun automatically, and upon operating the launch button by user will hit the target accurately. This feature helps locking the gun to target (target designation) by the help of smart intelligent system based on a micro controller or an ASIC or an FPGA. Gun drifting calibration and its correction factors are estimated and computed and corrected by the automated system, under the assistance of micro controller itself. A self-test / BITE calibration for gun stabilization health, number of ammunition rounds expended, and rounds left-over / available for firing etc., count will be generated and made available on the display to the user.

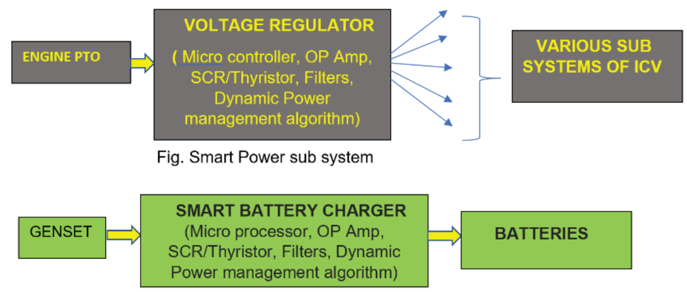

Figure 4. Smart power supply system.

3.3 Communication

Traditional communication sub system of ICV has two types, one for internal / intra communication, and other is for external communication. Internal communication within ICV can be controlled by the microcontroller-based design approach. Whereas, external communication can be implemented through FPGA based to facilitate smooth communication with various external heterogeneous communication systems. The spread spectrum and fast frequency hopping CDMA based digital radio sets deployed in the ICV used for intercommunicating with various types / variants of the external fleet of ICV in vicinity through an electronic bridge implemented with FPGA. This approach is more useful under relay-re-transmission mode for extended long-distance communication under the 4-tier mode. Which can link various ICVs' or variants through Wireless Sensor Network based approach. This smart communication

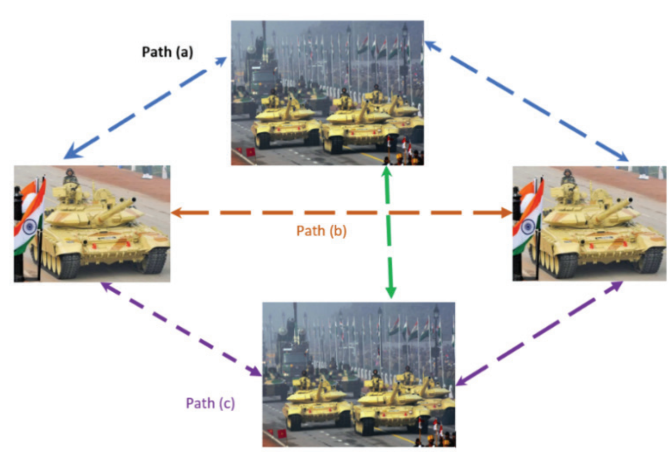

Figure 5. Networking for intercommunication of ICVs' (images courtesy: Republic day parade photos in open domain).

system facilitates the data / image / pdf / document / voice transmission, as well as in digital encrypted form.

A typical mechanism of interlinking of ICVs' is given in Fig. 5. The smart communication system also ensures the powering-on of external communication, only when some audio transmission takes place. This is achieved by the help of audio sensors, through software based / hardware-based FSM implementation, results in reduced power consumption.

3.4 Firing System (ATGM & Gun)

Firing system comprising, missile launching management, small-gun firing mechanism, takes the co-ordinates of the target obtained through optical sight or imagery system or remote sensing systems, that facilitates user to launch the ATGM and smaller gun's ammunition pointing at the target firmly and accurately. The missile health state including its internal battery voltage level is monitored through built-in micro controller of the missile. Missile co-ordinates reception into the ICV, missile's trajectory, guidance, velocity and acceleration control implemented in miniature / micro controller system or FPGA system, enable smart firing control system.

The reception of target coordinates and target image and characteristics, and its movement / velocity, and direction of movement etc., details from a satellite in real-time manner enables the ICV, for ease of bridging them with a hard processor implemented in PL section of FPGA. This helps the FPGA to coordinate the measuring ICV movement / its velocity and direction, for calculating the angle of fire, and automatically aligning the gun / ATGM firing system for the specific azimuth and elevation, and gives control signal to stabilizer system. Further the FPGA / micro controller generates the control signals to align the firing system precisely in optimal time for enabling the ATGM firing ready a self-test / BITE calibration will also cater for automated health checks of missile launcher, number of rounds expended, and rounds left over etc.

The proposed smart firing system implemented with a back-end computer / FPGA / micro controller having sufficient computational power, memory size, and effective computing algorithms implemented with the help of latest integrated-development-environment (IDE) and programming language. The computing device obtains the information from the Target tracking sensors viz. a SAR or laser or optical device or a digital camera, compares with database or by means of pattern classifiers, also estimates the target velocity, takes weather sensor data viz. speed-radar or Doppler-radar or wind-profiling-sensor, computes the co-ordinates of the target with respect to ICV itself and extrapolates the projectile at which it needs to be fired to hit the target. Based on this processed data the gunner, commander operates or fires the small gun or missile in optimal manner results in maximum damage and complete destruction of the target.

Degree or rate of fuel burning control mechanism, especially nozzle control mechanism also possible through microcontroller-based mechanism or by the FPGA based design alternately. Automated locking mechanism and firing to target is facilitated, after obtaining the co-ordinates of the target through imagery system / wire guided or laser guided sensory system help.

3.5 Navigation

The homing calibration of the navigation system of ICV are enabled with self-calibration through joint test action group (JTAG) boundary scan or built in self-test (BITE) test. ICV's co-ordinates and functionality or health are monitored through smart and intelligent micro controller-based navigation system as shown in Fig. 6. With the help of ICV movement in the tactical-battle-area (TBA), the information is captured in the form of map that can be stored in ICV's storage unit. This TBA map data is crucial for planning and deployment of striking force in the exigencies. Sharing this maps data between peer ICV helps for reference and correlation and their synchronization with each other. Navigational guidance support of NavIC to ICV fitted with ATGM is given in Fig. 6 for illustration.

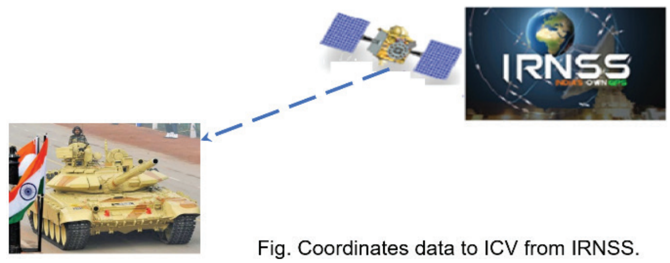

Figure 6. IRNSS / NavIC-satellite assisted navigation. (Image courtesy Indian Express newspaper, and Ministry of Defence).

3.6 Turret & Hull Electrical System

Hull is the bottom portion of ICV which safely holds all electrical wiring and its routing. Turret is the rotating top portion that seats above the hull in the ICV. Hull transfers electrical and various control signals to turret. It is also proposed to have local microwave-sensors fitted on ICV to achieve data fusion with satellite / drone borne data to further enhance reliability and accuracy of ATGM firing. The smart firing system comprises of missile launching management, gun firing mechanism that gets co-ordinates of the target after data fusion. This facilitates user to launch the ATGM missile and also small-gun's ammunition pointing at the target firmly and accurately. Missile health-status including internal battery voltage levels are monitored through built-in micro-controller of the ATGM and is made available to operator on his dashboard.

4. INNOVATIVE DESIGN OF ICV OPERATIONAL MANAGEMENT

New design approach with low-cost Micro controller-based interconnection of Smart Sub Systems described in Section II will enable an efficient design. Smart sub systems will make the traditional ICV into Smart ICV.

4.1 FPGA / SOC-IC based Smart Sub Systems of ICV

Alternate innovative design with SOC-IC FPGAs for interconnecting all Smart sub systems and its effective management synchronizes various operations parallelly and optimally. In this approach, processing Sensor data, and measurement & control signal generation and the synchronization with state machines implemented in FPGA based SOC-ICs. This results in re-configurability and ease in maintenance. The FPGA based proposed design is given in the Fig. 7.

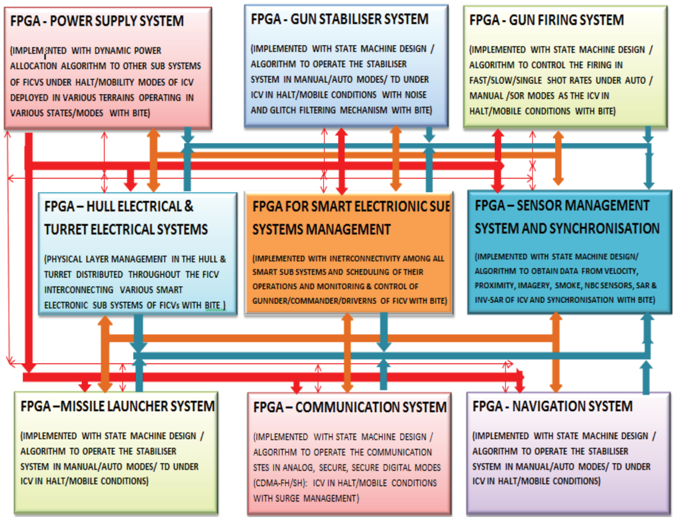

Figure 7. FPGA – SOC based advanced systems.

Legend:

SCHEDULING AND SYNCHRONISATION
SENSOR SIGNAL
POWER SIGNAL
POWER SUPPLY

Due to increase in synchronisation and time efficient, and increased accuracy of internal operation of the each sub-system of the ICV, and increase in navigational support using NavIC Satellite system, and increase in accurate assessment of the object characteristics, increases the confidence level beyond 95 % and thus increases mission time MTBF . This reduces the Mission time failure rate. With the increase in visibility / failure identification of sub system of the ICV or ATGM on the user / operator's control panel during Built-In-Self-Test activity, increases the precision of replacement of faulty sub-system or faulty card or faulty component witin very short span of time. This contributes for reduced recovery time / MTTR to almost nigligible. This contributes for overall increase in Reliability, with the proposed approach.

4.2 Micro Controller Based Interconnection of Smart Sub Systems of ICV with Serial Bus

Sensors to sense the mechanical or electrical parameters and to convert these parameters to electrical signals viz. sensors for seeing through all weather conditions. IRNSS / NavIC for location (homing), temperature sensors, oxygen / smoke sensors, velocity sensors, and pressure sensors. Signal conditioning mechanism to convert the sensor output to scaled electrical means (i.e., parameters viz. current, voltage, frequency, pulses, and their length). conditioning of these electrical parameters, and interfacing to a processor that is equipped with ports, memory, display, control console (wherever controlling is necessary) (alternately, a micro controller), and its software programs (both, for control signal generation and sensor data processing).

Electronic systems of ICV and their operations can be organized under various environmental conditions of end user (i.e., modes like different terrain selection desert, rocky, mossy, sandy, hilly terrain, in water bodies, various gradient / cross country, in various gears or speeds; auto, manual, semi-auto, in nuclear biological environments etc., through mode or configuration selection data files) through software or hardware based finite-state-machines (FSMs). Central processing unit (CPU) implemented using FPGA will make the system as re-configurable, provides agility, flexibility, and adoptability. Wherein micro controller-based implementation of certain sub components of the smart electronic sub systems will facilitate simple and easy design.

Simplest automation architecture of the Vehicles is by using micro controllers, and interfacing them with I^2C and CAN controller protocols. In addition, few accessory circuitries, or

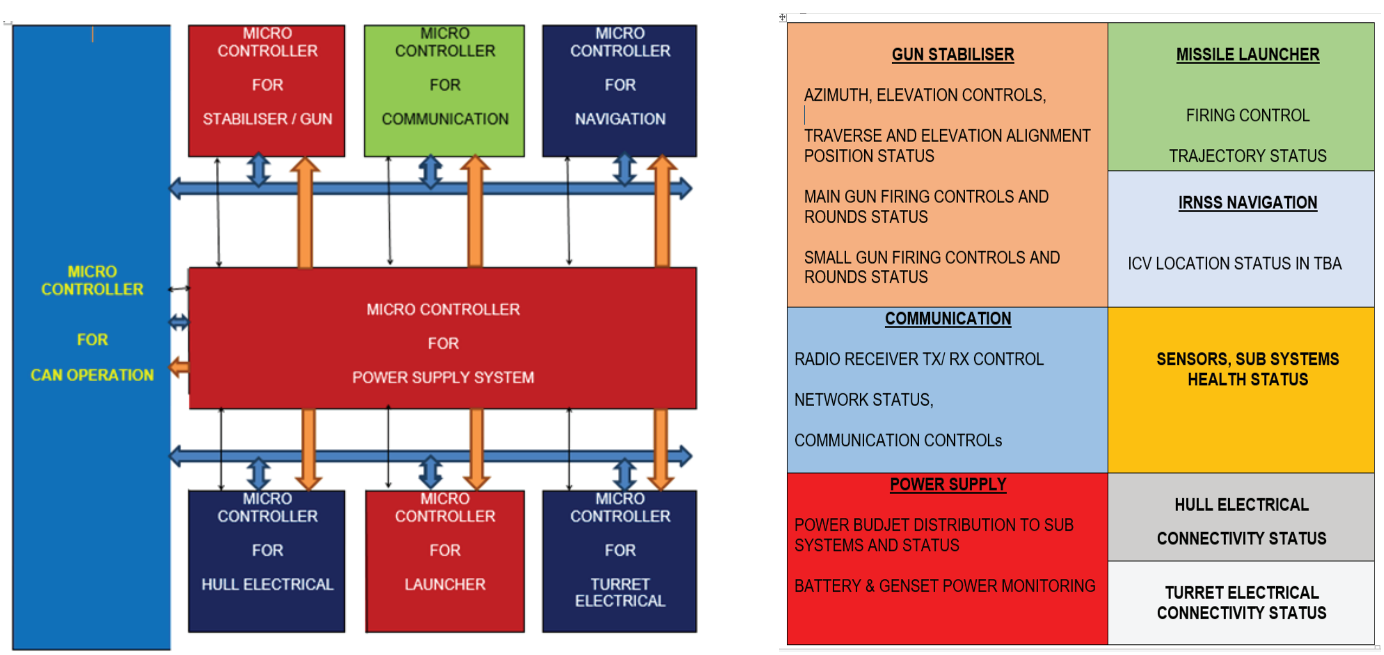

Figure 8. Interconnection of smart sub systems of ICV with CAN architecture for serial bus.

components like analog to digital converter (ADC), digital to analog converter (DAC), multiplexer (MUX), de-multiplexer (De-MUX), buffer amplifiers, etc., implemented through FPGA (single ASIC under mass production). Proposed schematic design of interfacing the electronic sub systems using CAN (Controlled Area Networking) controller architecture of ICV / variants is depicted in Fig. 8.

4.3 Inter Communication Between ICVs' Using Data Communication Beyond Visual Range

Each variant of ICV incorporated with hard coded IMEI number and interconnected connected with TCP-IP model with advanced Software Defied Radio or alternately with HF / VHF radio sets fitted in each ICV brings with VOIP.

ICV incorporated with secured and encrypted HF and VHF Radio transmission link is used for external communication medium, amongst other ICV can be interconnected through Network based system results advantage of networked ICVs'. This extends a wireless Sensor Networked ICVs' at various levels scaled up from Regt level, to Brig, Div and Corps level with upward connectable / compatible. The intercommunication in the networked ICVs' can be done in a manner depth first, breadth first or such state-of-the-art advanced approach. This also helps sharing the live data exchange amongst peers i.e., source to destination ICV, with the hardware systems are loaded with Linux / Unix operating system components. A typical interconnection of ICVs' is described in Fig. 9

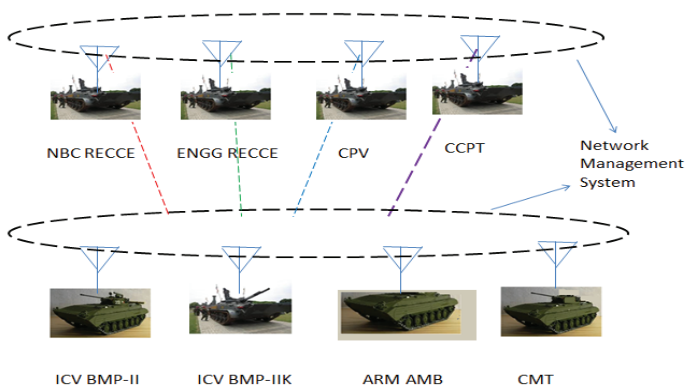

Figure 9. Networking of ICV and variants.

This ensures the safety of the cyber-physical-systems (CPS) of the ICVs'. Also enables the fleet of ICV for their integration with C4I systems, for optimal and synchronized use as per the user / operator's choice. This interconnection for intercommunication of ICV enables Machine-to-Machine technology-based ICV. This helps identifying the nearby adversary easily and also helps for operator / user with increased redundancy of safety during operations. This approach enables intercommunication between ICVs', extended in the cross country and physically separated beyond visual range benefits the user largely.

5. NAVIC AND SMART-SMALL-SATELLITE / DRONE SUPPORT FOR DIRECT DELIVERY OF IMAGES FOR NEAR REAL-TIME AUTONOMOUS FIRING OF NG-ATGM IN TBA

In traditional scenario globally, conventional offline maps of the TBA are compiled in advance. However, in a real-world scenario, actual ground realities can change very dynamically within a very short time. This necessitates the driver / user to have knowledge about the latest updated near-real-time TBA information like a culvert or elephant-holes, underground-tunnels, broken-bridges etc.

Research-works have progressed for near real-time image-processing[18-20] which are encouraging for direct delivery of images to the end-user. Research works progressing in field of delivery of the terrain / road condition to user is also encouraging. This satellite-based final processed images transferred to an ICV enabled with real-time direct classified images of the TBA, target & its characteristics turns them into real-time battle-field intelligence. Further providing the traditional manner of way points in an offline mode sometimes make the scenario belated, that results of-late. Overcoming such vulnerable situation is possible, in case a satellite-based object tracking and providing the velocity, direction, of movement of target and its' real-time locational co-ordinates to driver. This will enable the driver to move ICV in a semi-autonomous mode in an optimal time to reach the required location.

Hard-processor implemented in the "PL" of FPGA / in the computer is programmed to generate the firing action.

This approach can be extended to upgrade outlived ICVs with NavIC and satellite / drone based imaging and direct delivery of the target information to the ICV platform. This can also help the outlived ICV to be re-used for present day TBA requirements.

A computer system fitted in the ICV (an FPGA based processor in (PL), process the satellite data, and generates the control signals to aligning the firing system, in azimuth, elevation. This alignment of firing system, taking into consideration of the parameters viz. velocity, direction of movement in 3D space of TBA, the target coordinates, wind-speed and wind-direction and other environmental parameters that have bearing on the firing-angle alignment.

This will enable either the new-ICV or the outlived-ICV that has been refurbished as cost effective firing system-design, for effective / accurate use at TBA. A typical smart ICV enabled with target characterization by radar sensor and direct delivery to end-user, for effective / smart NG-ATGM firing, is given in Fig. 10.

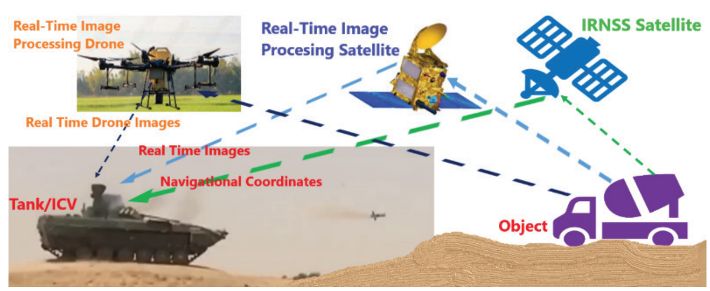

Figure 10. Drone & satellite processed real-time image support and NavIC support for ICV.

(Drone Image, Courtesy @ https://govtschemes.org/namo-drone-didi-yojana-2025/ , Small Satellite image courtesy @ ISRO website, Tank image Courtesy @ https://www.youtube.com/watch?v=4odMvv__Fak)

Research works are also progressing in the field of AI-ML / Deep learning systems giving scope for our proposed approach for delivery of terrain / road condition to the user, i.e., the navigational guidance support can be provided to smart ICV along with direct delivery of object images18-20.

6. AI - ML APPROACH FOR ENHANCED AUTOMATION TO IMPROVE FIRING-RATE OF ATGMS

An artificial neural network (ANN) is a type of complex computing system that models individual neural connections that mimics the information processing abilities similar to that of a human-brain. ANNs for accurate estimation of rain-fall accumulation was implemented in the works of[27], considers and discusses various input parameters contributing for respective estimations / computation with inner layers and gives the outputs at Output layer. Research works of authors[28] examines object detection using Deep-Learning, CNNs. Convolutional neural networks (CNNs) are effective in resolving complex computations associated with inter-related multiple input parameters.

Research works of authors[29] examine solving real-time problems specifically the of hardware chip designs and their performance suitability for upward scalable computations. Authors conduct study on field-programmable gate arrays (FPGAs) in chip logic verification, utilizing parallel computing and pipelined design principles to enhance execution efficiency. Authors conclude that CNN chip has demonstrated the optimal delay of 8.750 ns, while operating at 552.00 MHz frequency, with 98.78 % accuracy with a scalable design of 64 neurons processing[30]. Research works of[30] develops a Deep Learning Framework for Indian Road Network Extraction from Cartosat-2E Satellite Images model achieved, an accuracy of 94.15 %, precision of 82.42 %. This gives inputs for future accurate estimation model using NavIC system.

The ongoing research work gives scope for extending the convolutional-neural-network (CNN) based solution for design of automated testing and recording the test parameters of various production stages and also during QA phase of ATGMs. This approach extended to conduct of functional-parameter-tests of ATGMs at TBA, helps for assured confidence before their operational use. This method of capturing various test parameters helps in overcoming errors due to traditional / manual way of testing. This helps to make it easy for the operator / user during ATGM firing operations. Various types of control approach i.e., supervised, un-supervised, and hybrid (combination of both) are possible for adoption in designing such AI-ML based ATGM firing system.

The principles of AI-ML in CNN / ANN[27-29] can be applied for improved success and autonomous ATGM firing from ICV taking into consideration of road conditions[30]. Direct delivery of satellite / drone-based object images support[18-20] of the TBA region to the operator / user in ICV is proposed in this sub-section.

Data about previous and present firings and corresponding situations, and environmental parameters under which the ATGM firings takes place, similar to the data[31] are captured, with block chain enabled manner, builds a repository of data & their relationships of ATGM firings that help to design an accurate Indigenous CNN / ANN[27-28]. The proposed method of new-Indigenous CNN / ANN aids and enhances[32] for future use. This helps for assessing the test results at various production stages that they are complying to the pre-designed/ set-rules are falling within limits/acceptance criteria. Aiding this data repository with un-supervised or a set of rule-based approach helps design of robust hybrid system. “Programming Logic” (PL) section of FPGA helps AI-ML or a CNN based data-engine runs in an efficient manner in the ICV.

This enables for control of azimuth and elevation motors and drives for their accurate/precise movements withing shortest times and helps alignment of the firing system in an

Table 1. Role and benefits of CNN in ICV

Step	Input data	Cnn role	Output / benefit
Target detection	High-res eo/ir imagery from icv optics + satellite imagery	Cnn detects shapes, rcs signatures, and movement patterns of enemy armor, adversary teams, uavs	Accurate classification of threats with >95 % confidence
Target tracking	Continuous navic satellite data feed + ground-based radar sensor data	Cnn predicts target movement trajectory in real-time, with the data input received from navic satellite-based co-ordinates as well as real-time satellite images generated by cp-sar payload for target.	Ensures missile aim-point is always on target, with the help of automated tuning of azimuth and elevation angle motor drives for aligning the atgm launcher
Obstacle / aps detection	Optical + radar inputs	Cnn detects enemy active protection systems (aps) & obstacles	Adjusts missile approach path to avoid interception, and gives input for the steering control, for left or right, forward or reverse movement signals by fpga hard processor.
Firing solution generation	Ballistic parameters + navic/ ins + weather data	Cnn fuses data for optimal firing angle of azimuth & elevation, lead, and timing for firing	Higher probability of first-shot kill
Mid-course updates	Satellite relay + onboard sensors in atgm	Cnn processes new data, corrects missile path and updates the alignment of the optical sight cursor of the icv	Compensates for target movement or evasive action
Post-impact assessment	Battle damage imagery	Cnn verifies hit and assesses damage	Immediate re-engagement if needed

automated manner. This results in ease of firing of the ATGM fitted in the ICV and also from various other platforms.

Designing such system with the real-time images directly from remote sensing satellites and target coordinates support from NavIC helps for safe & optimal navigation of the ICV fitted with ATGM moving towards target.

In future this can help towards the platform movement in an automated / driverless manner with redundancy of safety by accompanied support of driver, commander, and gunner for ensuring safety, during exigency.

Use of neural networks will enable more accuracy of test results and provides accuracy in quality of sub systems, and firing which results in the increased mean-time-between-failures (MTBF). Neural-networks based fault analysis / BITE checks enables the user to easily locate the faulty sub system and its replacement becomes easy. This helps for reduced Mean-Time-To-Repair (MTTR). Table 1 brings out typical benefits of AI - ML and CNN based approach & and its benefits for its use in ICV.

7. CONCLUSION

ATGMs are primarily fired from ICV platforms. Research is progressing with focus on ease of user's / operator's requirements towards futuristic NG-ATGMs and also for improved success-rate of firing from a smart ICV and Tanks with capabilities for target detection & identification with near real-time satellite images and navigational guidance with NavIC. Advances in AI-ML enable the smart ICV to move precisely to a particular location within improved timelines for faster firing of AGTMs.

These new requirements at the TBA are needing the aspects viz., high-impact-kill power, optimum use of propellant, maximum range cover, high-altitude applications, high-velocity (high-Mach) / shorter flight-time, high-precision, high-accuracy, mission-time reliability, fire and forget, local / global navigation guidance, cheaper costs, connectivity with M2M smart systems. These requirements compel quality as essential attribute that need to be consistently maintained from the initial stage of ATGM design, development stage onwards, till prototyping / its evaluation and in bulk production. QA aspects applied to the smart ICV with support from NavIC constellation and satellite / drone images.

Proposed approach / framework model gives benefits in multitude to various stake holders. It helps the designer, the design with re-configurability, agility, Fine course corrections during prototyping stage, ease in upgrading the design to newer technology or newer trends. Proposed design helps production agency, with ease in assembling at the assembly line / floor. Also helps with ease of fault analysis. The proposed approach helps the QA agency with measurement & repository of test results, ease of testing, reduced testing hours, reduced manpower requirements for testing. Brings ease in carrying process-audit / quality-audit. The proposed approach helps the maintenance agency, with ease in fault identification, ease in defect analysis (since test logs are readily available in the memory storage of the smart electronic sub systems).

The health of the ICV is available on the touch-screen display which makes it easy for the user in TBA. New innovative approach for interlinking of ICVs' for beyond visual range communication has also been described. New Generation ATGMs & for improved reliable firing with the support of Neural Networks have been discussed, which implicitly includes Industry 4.0 / 5.0 from of design till production processes. Benefits of the proposed novel approach to all the stakeholders has also been discussed.

Potential future scope of work is described in succeeding paras.

Our research work comprising an innovative approach proposed for NG-ATGMs firing from smart ICVs can also be extended in future for firing from smart tank platforms. The proposed innovative approach is extendable for improved successful firing from other smart platforms like truck or from man-portable ground launcher etc.

Further research work can include the navigational support from NavIC and real-time images support from satellites / drones which can extend beyond visual range such as 50 km or above of the 5th gen-ATGM. This extended research-work gives a new dimension of sensor-to-shooter scenario for its use in NG-ATGM firing in future.

The AI-ML approach proposed in this research-work can be extended as future work with reinforcement learning. The effectiveness of improvement in firing of the NG-ATGMs noted with the proposed innovative approach can be studied, analysed and can be well correlated for internal statistical purposes of the user. This also helps the respective stakeholders for review at regular intervals, so that the ANNs / CNNs can also be reinforced and improved for obtaining more precise firing results over the period of exploitation.

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